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ABSTRACT

Explainability is a key aspect of machine learning, necessary for ensuring transparency and trust in decision-making processes. [...]

Keywords: Artificial Intelligence, Black Box Models, Cosine Similarity, Deep Learning, Explainability, Feedforward Neural Network, Logic, Regularization, Rule Extraction

1. INTRODUCTION

Artificial deep neural networks (DNNs) have been empirically shown to have high predictive accuracy and perform well on various machine learning problems. [...]

2. EXPLAINABILITY

Explainability provides the foundation for interpreting how models generate outputs and identifying the reasoning behind predictions. [...]

3. COLOSSUS

3.1. Context Analysis

The study examines how cosine similarity can be leveraged to analyze neural weights and biases. [...]

3.2. Scoped

The scope of the COLOSSUS framework focuses on feedforward networks with fully connected layers. [...]

3.3. Proposed Method

The proposed method integrates cosine similarity and first-order logic (FOL) to generate interpretable rules. [...]

4. LIMITATIONS AND FUTURE WORK

Future research will explore extending the model to recurrent or convolutional networks. [...]

5. RELATED WORK

Several prior studies have explored rule extraction from neural networks, including decision-tree surrogates and symbolic regression methods. [...]

6. MAIN CONTRIBUTIONS

This study contributes (1) a hybrid symbolic–numeric rule extraction algorithm, and (2) an interpretability metric using cosine similarity and entropy. [...]

7. CONCLUSION

The integration of cosine similarity and logic-based reasoning enhances the explainability of trained deep neural networks. [...]

CONFLICTS OF INTEREST

The authors declare no conflict of interest.

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